BULACAN POLYTECHNIC COLLEGE
Bulihan, City of Malolos, Bulacan
BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

DueWise: Design and Development of an Assignment and Task Management System for Students

In Partial Fulfillment of the Requirements for the Degree of Bachelor of

Science in Information System

Submitted by:

BSIS 2C (Group 1)

Alab, Sopia C.

Billalba, Lexter John M.

Maron, Renz Ryan G.

Morandarte, Mark B.

Salvador, Cedrick John

October 2026

I. Project Overview

Project Description

A mobile application for managing the tasks assigned to students at school, and for organizing their school work, called DueWise. Pupils are sometimes asked to complete multiple assignments across a variety of subjects with various requirements and deadlines. Managing these tasks can be hard, particularly if several assignments must be finished in one week. DueWise provides one place where students can manage their assignments, monitor their progress, and access tools that can help them with their schoolwork.

It allows users to add, edit, view, and delete assignments. Each task can include a title, description, subject, due date, priority, and status. Students can update the status of their assignments and mark tasks as completed when they finish them. These features help users keep their task information organized and easily check which assignments still need their attention.

DueWise also includes a dashboard that gives users an overview of their academic tasks. It displays information about their assignments, including pending, completed, and overdue tasks, along with their overall progress. Through this feature, students can check their current workload without opening each assignment individually. The application also provides reminders when deadlines are approaching, helping users keep track of tasks that need to be completed or submitted on time.

Another feature of DueWise is its camera and scanning function. Users can open the camera and choose between scanning a paper or scanning a math problem. When scanning a paper, the application can convert it into a digital file, such as PDF, DOCX, PNG, and JPEG, which users can save or download.

If the scan is unclear, they can scan the paper again. For math problems, the application can process the scanned problem, solve it automatically, and display the answer. These features give students additional tools they can use while working on their academic requirements.

Designed for mobile devices, DueWise has a user interface with sections for the Dashboard, My Tasks, Progress, Notifications, and Profile. These sections allow users to access the application's main features and manage their assignments in one place. For students to be able to access their own accounts, the application will also feature account registration and login. It's designed to improve the understanding and ease of use of the task management and navigation processes.

In general, DueWise will enable students to handle their academic workloads with a single mobile application, encompassing assignment management, progress tracking, deadline alerts and scanning capabilities. This provides students with an easy-to-use tool for managing school work, tracking their progress and accessing extra resources when doing school tasks.

Problem Statement

Students often struggle to handle multiple academic tasks and assignments. They can miss deadlines for submissions, find it hard to decide which tasks to work on right away, and forget to check if there are unfulfilled requirements. These challenges may cause students to submit late, have difficulty managing their time, and have too many academic tasks.

Target Users

It is designed for students who perform a variety of activities, projects, and requirements in various subjects and want to have a mechanism for organizing their schoolwork. It can be used by students who want to keep track of their academic responsibilities and complete their assignments on time.

Proposed Solution

The proposed solution will be to create a digital platform for students to handle their academic work from a single platform. It will be easier for the student to keep track of school needs as the system will be organized by subject, deadline, and priority. It will also give a student a list of pending, in progress, and completed tasks, making it easy for them to identify which tasks still require them to complete. This is a system for maintaining an orderly approach to school assignments and classwork.

II. Objectives, Scope and Features

General objective

The main purpose of this project is to make an application which consists of a mobile-based Assignment Task Management Application that will assist the students to manage their assignments, track their task processes and the deadline of assignments, and scan a paper-based document and convert it into a digital document and scan math problems in a secure and user-friendly platform.

Specific Objectives

The main objective of DueWise is to help students manage their academic tasks through a mobile-based assignment task management application. Specifically, it aims to:

1. Allow students to create, view, edit, and delete assignments by providing details such as title, description, subject, due date, priority, and status.
2. Provide a dashboard that displays assignment information, including pending, completed, and overdue tasks, to help students monitor their academic workload.
3. Enable students to update assignment status and monitor their progress in completing school requirements.
4. Provide reminders and notifications for approaching deadlines to help students keep track of their assignments.
5. Develop a paper scanning feature that allows students to scan documents, convert them into digital files, and save or download the files. It also allows users to rescan papers when the scanned image is unclear.
6. Develop a math scanning feature that allows students to scan math problems, process them automatically, and display the answers.
7. Provide a login and registration system that allows students to access their accounts and manage their assignments.

Scope and limitations

Scope

This project will primarily be based on the creation of a mobile app named DueWise to facilitate students' organization and management of their studies. It functions on the user interface, account management, assignment management, task status monitoring, progress tracking and notification about deadlines. Students can manage their pending, completed and overdue assignments, create, view, edit

and delete assignments. The scope also includes a scanner feature that allows users to scan paper documents, convert them into digital files, save or download the files, and rescan unclear documents. In addition, the math scanner allows users to scan math problems, process them automatically, and display the answers. These features are intended to help students manage their schoolwork and access useful tools for their academic tasks through one mobile application.

Limitations

DueWise will not be the school's official learning management system (LMS). It does not support direct submission of assignments to teachers and automatic calculations of grades for students. The usefulness of the application also relies on what information it delivers by its users. The information may be inaccurate if the task is incorrectly entered, or the deadline is set incorrectly. In addition, the App offers students ways to track what is expected from them, but it does not ensure they finish or submit it on time.

Major Features

- **Task details** - Shows the full information of a task, including its title, description, subject, due date, priority, and status. Students can edit the task, update its status, mark it as completed, or delete it.
- **Create/edit tasks** - Allows students to add a new task or update an existing one by entering the title, description, subject, due date, and priority level (Low, Medium, or High).
- **My assignments** - Displays the list of all tasks with a search bar, filters (All, Pending, In Progress, Completed), and sorting options. Priority levels are color-coded, so urgent tasks are easy to spot.
- **Progress Monitoring** - Shows the student's overall progress as a percentage, along with a breakdown of tasks by status (completed, pending, overdue) and by priority (high, medium, low).

- **Notification and reminders** - Alerts students about upcoming deadlines and overdue tasks and keeps them in a reminders list so they can stay on schedule.
- **Profile and settings** - Lets students view their account, manage notification preferences, and log out of the app.
- **Login and Sign Up** - Secure user authentication to register accounts, log in, and protect personal task data.
- **Scanner** - Allows students to scan paper documents and convert them into digital files, such as PDF, DOCX, PNG, and JPEG, which they can save or download. It also includes a math scanner that can automatically solve scanned math problems and display the answers. Students can rescan documents if the scan is unclear.

III. System Design

System/Application Architecture or Flow Diagram

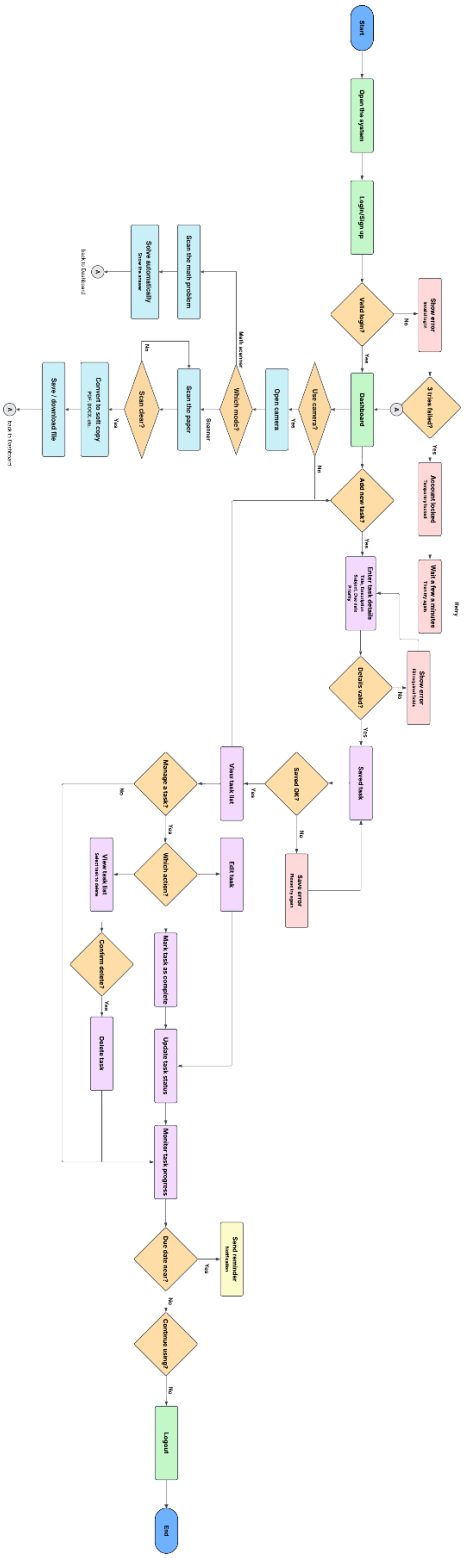


Figure 1.

The Assignment/Task Management System flowchart is used to show students how they utilize DueWise for school assignments. This action starts with the log in/sign-up of the students. The system verifies their login credentials and if they fail to log in three times the account is temporarily disabled. After Login, students are able to create new tasks with title, description, subject, due date, and priority. The system checks the information prior to saving the task. If the save is unsuccessful, the student will get an error message and be given a chance to save again.

Students are able to see the list of their tasks, edit the options for an assignment, delete a task after the confirmation, complete a task and oversee their status. DueWise also provides reminders when assignment deadlines are approaching.

Students can open the camera and choose between scanning a paper document or a math problem. Scanned papers can be converted into digital files, and saved or downloaded. If the scan is unclear, students can scan the paper again. For math problems, the system processes the scanned problem, solves it automatically, and displays the answer. After completing these actions, students can return to the dashboard and continue using the application or log out to end their session.

Use Case Diagram

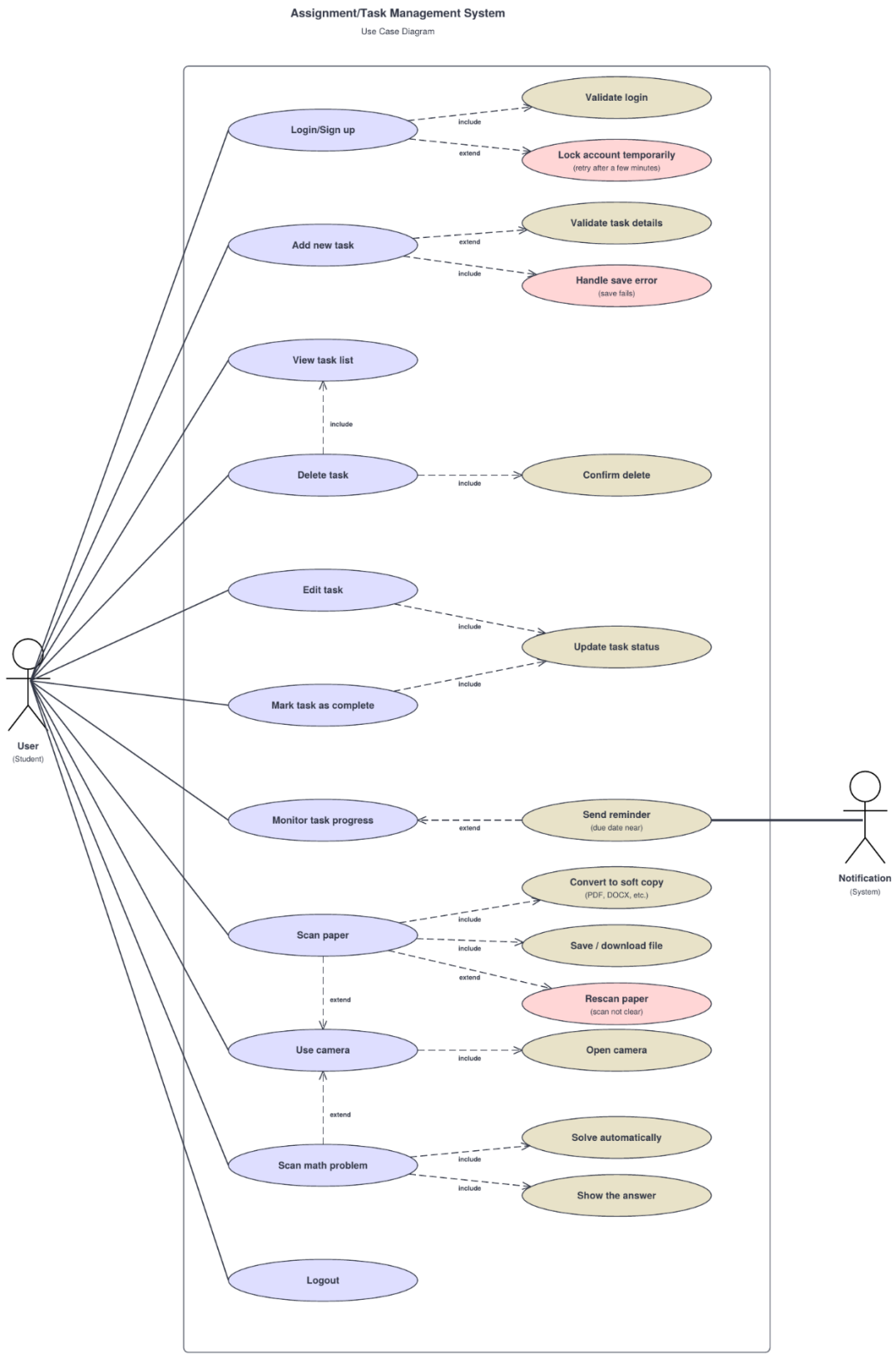


Figure 2.

The Use Case Diagram for the Assignment / Task Management System displays how students will interact with DueWise to handle their schoolwork. The student can register and log in, add new tasks, view their tasks list, edit or delete tasks, mark a task as complete, and track their task status. The system also includes login validation and temporarily locks an account after three failed login attempts. On adding a Task, the system verifies the Task entered before saving and is able to show an error when it doesn't save correctly. It also provides reminders whenever there are assignment deadlines.

In addition to task management, the use case diagram includes a scanner feature. Students can use the camera to scan paper documents and convert them into digital files, which they can save or download. If a scan is unclear, they can scan the paper again. The math scanner allows students to scan math problems, process them automatically, and view the answers.

The diagram uses include relationships for required processes and extend relationships for additional actions that occur under specific conditions. Overall, the system is designed to help students organize their assignments, monitor their progress, keep track of deadlines, and access scanning tools for their academic needs.

ER-Diagram

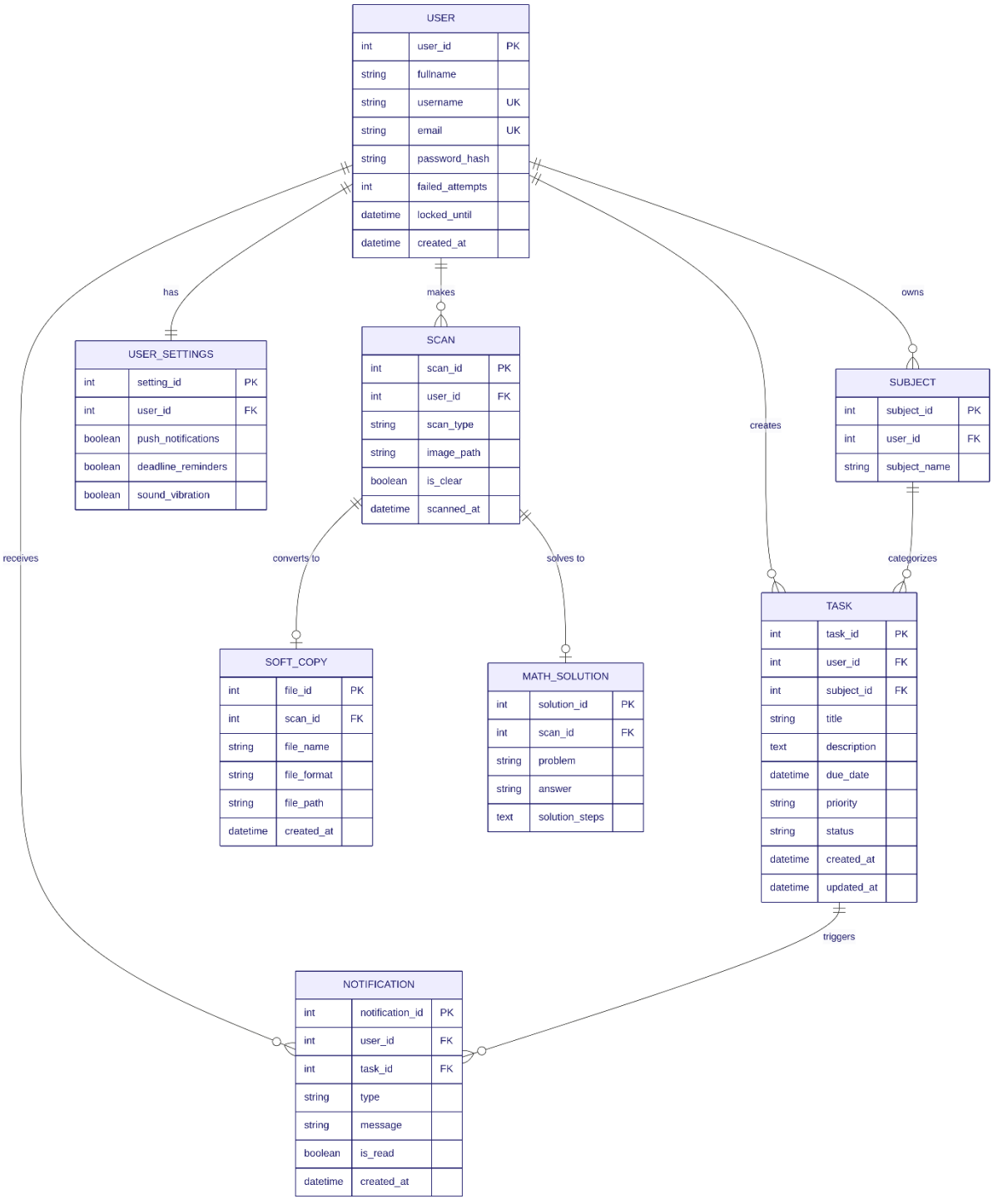


Figure 3.

IV. UI/UX and Implementation

UI/UX design

This document is describing the visual system, screens, navigation and new Scanner feature (Document and Math modes) of DueWise, a mobile application for tracking tasks and deadlines for students. DueWise uses a navy and indigo palette on a light page background, set in Roboto, so every screen reads as one system.

Token	Value	Used for
Navy	#1E2250	Status bar, page header, primary buttons
Indigo	#303781	Frame background, active accents
Page background	#EFF3F4	Content area behind cards
Border	#CED3D8	Card and input outlines (1 px)
Ink	#12151C	Body text
Error red	#BF292D	Delete, Logout, error states
Stat tints	#FFF9E9, #F0FAF3, #FCF0ED	Dashboard stat cards (Pending, Completed, Overdue)

Typography and Layout

It has adopted Roboto typography in its user interface, which makes for clear and consistent text. The information is easy to read due to different text sizes used for titles, body text, labels, and navigation. It is organized by sections and there is good spacing used in the layout. It features a top header, scrolling content area and a navigation bar at the bottom that provides access to the various areas of the application. Rounded corners are also implemented to produce an easier-looking user interface.

Navigation and user flow

There are five bottom-navigation tabs (Home, My Tasks, Progress, Notifications, Profile) which users switch between, and the Scanner can only be accessed from the bottom-navigation button "+" on the Dashboard. The Scanner is initially in Document mode and the toggle is set to Math. If you scan the object and get a blur, you can press "Retry," and if you scan the object later and get "Math Solution," you can press "Scan Another". The other Add menu option Create New Task follows the one already added to the flow of tasks.

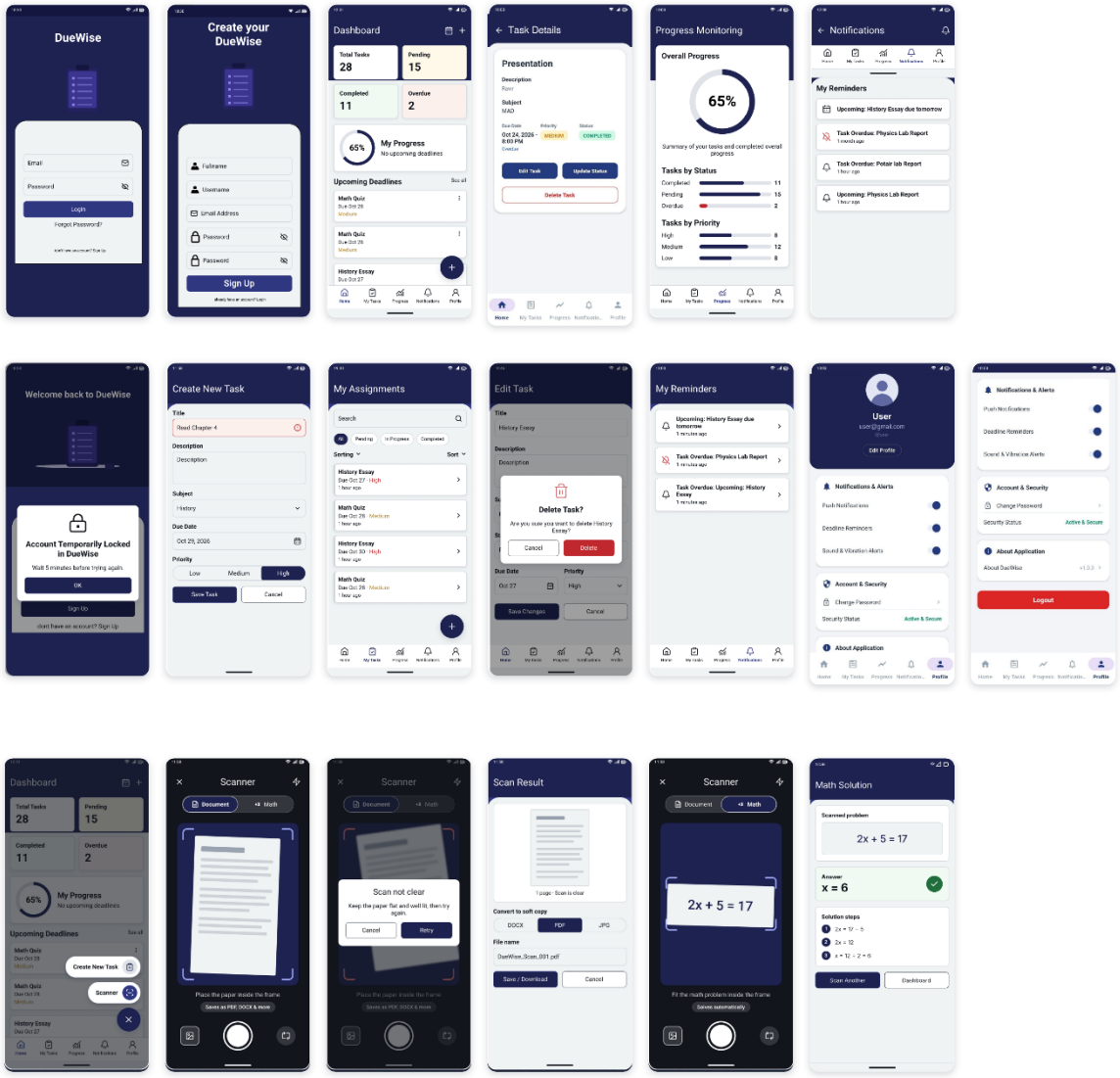


Figure 4.

Development tools

- **Figma** Used to design the User Interface and create the visual layout of the application before coding. It helps us plan the app's appearance, screen components, colors, typography, and user navigation flow. With Figma, our team could easily collaborate and preview the design to make sure everything looks good before actual development begins.
- **GitHub** To store and manage our project code securely as a version control system and to let the team work together on the same files, track changes, revert when needed, and keep a reliable backup.
- **Android Studio** To serve as our main workspace for writing, editing, and organizing the application code, and to provide all necessary tools for testing, debugging, and building the final installable version of the app.

Technologies used

- **Kotlin** To serve as the primary programming language that builds the logic and functionality of the app and to write modern, concise, and clean code that works smoothly with Android and has fewer errors.
- **Gradle** To manage project dependencies, libraries, and build configurations automatically. It organizes and manages external libraries and resources needed for the app to run properly.
- **Kotlin DSL (.kts)** It provides a cleaner way to write our project configuration files using familiar Kotlin syntax and to make the build configuration easier to read, understand, and maintain since our team already uses Kotlin.
- **XML or Jetpack Compose** XML is the standard and widely used way to design app layouts, while Jetpack Compose is the newer, modern toolkit that lets us build the user interface more simply and efficiently. It defines the overall design and structure of the app interface, ensuring it looks good on different device screens.

Major application screens

The major application screens of DueWise show the different sections that users can access while using the application. Each screen has its own purpose, allowing students to manage their assignments and access the available features easily.

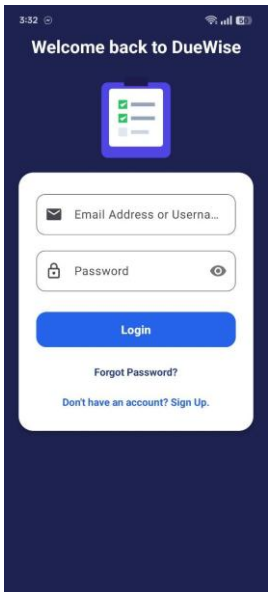


Figure 5. Login

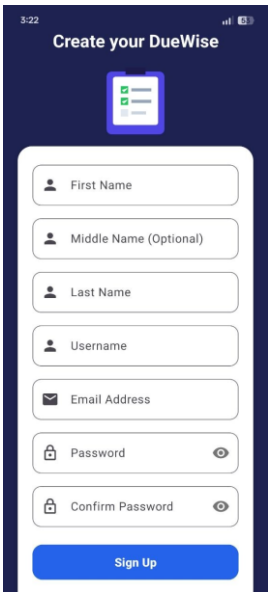


Figure 6. Sign Up

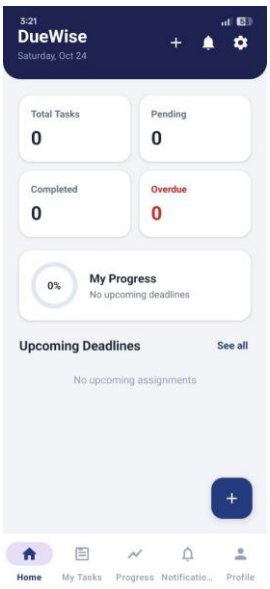


Figure 7. Dashboard

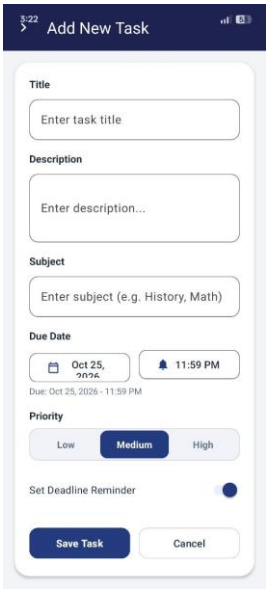


Figure 8. Add New Task

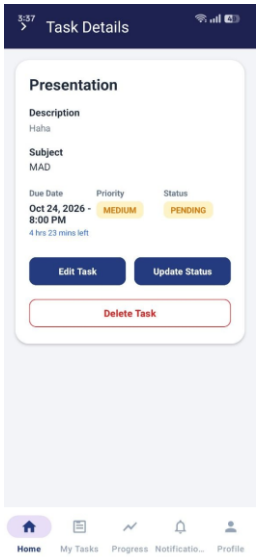


Figure 9. Task Details



Figure 10. Scanner



Figure 11. Soft Copy

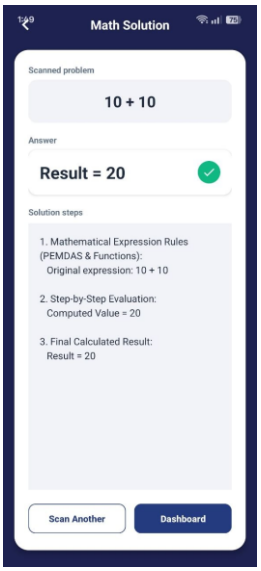


Figure 12. Math Solving

Feature implementation

DueWise was developed to help students manage their academic tasks in one application. It includes a user management system that allows students to register and log in to their accounts, while the dashboard provides an overview of their pending, completed, and overdue assignments. Users can create, view, edit, and delete assignments by entering details such as the title, subject, description, due date, and priority. They can also mark assignments as complete, monitor their progress, and receive reminders for approaching deadlines. Another feature is the scanner, which allows students to use the camera to scan paper documents, convert them into digital files, and save or download the files. If a scanned document is unclear, users can scan it again. Students can also scan math problems and receive automatic processing and display of the answers with the math scanner. There are also Profile and Settings pages included in the app for users to view and adjust the settings, which they can access. These features allow students to manage assignments, track due dates, monitor progress, and scan support their learning.

V. Testing and Evaluation

Test cases	Results
Register an account	The registration function was successfully tested. The system allows new users to create an account by entering the required information. The account was successfully created and saved in the system.
Log in with valid credentials	The login function worked successfully. When users entered valid credentials, the system verified their information and redirected them to the dashboard, allowing them to access the application's features.
Add an assignment	The Add Assignment function was successfully tested. Users were able to enter assignment details, including the title, description,

	subject, due date, and priority. After saving, the assignment appeared on the task list.
Mark a task as completed	Mark as Completed function worked correctly. When users marked an assignment as completed, the system successfully updated its status to "Completed," allowing students to monitor their progress.
Edit an assignment	The Edit Assignment function was successfully tested. Users were able to modify existing assignment details, and the system saved and displayed the updated information correctly.
Delete an assignment	The Delete Assignment function worked successfully. Users were able to select and delete an assignment after confirmation. The selected assignment was removed from the task list.
Scanner	The Scanner function was successfully tested. Users were able to scan paper documents, convert them into digital files, and save or download the files. The system also allowed users to rescan unclear documents and scan math problems to display the answers.

VI. Conclusion and Recommendations

Project accomplishments

The development of DueWise: Assignment/Task Management System successfully achieved its main objective of helping students organize and manage their academic tasks efficiently. The application includes functional features such as account registration, login, adding assignments, viewing task lists, editing and deleting tasks, marking assignments as completed, and also the scanner. It also allows students to monitor their task progress and track upcoming deadlines. Based on the functional testing results, all tested functions operated successfully and produced the expected outcomes.

Problems encountered

Group faced a number of difficulties designing and developing the application to DueWise. One of the biggest challenges was designing the user interface so users could navigate the screens easily and intuitively. The group needed to come up with ways to manage assignment details like subject, assignment due date, and task status to allow users to manage their school requirements well.

Another challenge faced was ensuring the features in the application function as expected, particularly when creating, editing and marking assignments as pending or completed. The information on the dashboard was also structured so that users could easily understand what they were doing. The team had to think about what they would be using for their application data and how they would handle the storage and management of these data, to ensure that they are organized and accessible. Despite these difficulties, the group persisted in the design and development of DueWise to produce an application that would enable students to complete their academic work more efficiently.

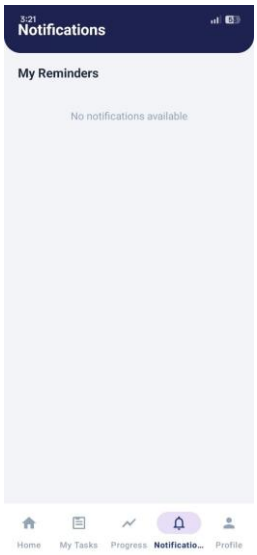
Recommendations/future improvements

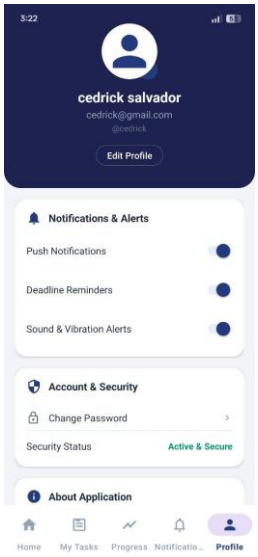
The UI of DueWise can be enhanced to create a more organized and attractive-looking application. The layout, color, buttons, icons and text styles can be enhanced to make it consistent on all screens. This will help students easily understand the information displayed in the application. Navigation can also be improved to make it easier for users to view and complete important features and move between screens. Menus and buttons can be arranged in a more organized manner and with an easy-to-understand layout so that a student can use the features without confusion. Finally, testing results and user feedback can be used continuously to improve DueWise. Regular updates allow problems to be identified and corrected, improvement of app performance, and to make app more reliable, accessible and useful to students to manage their assignments.

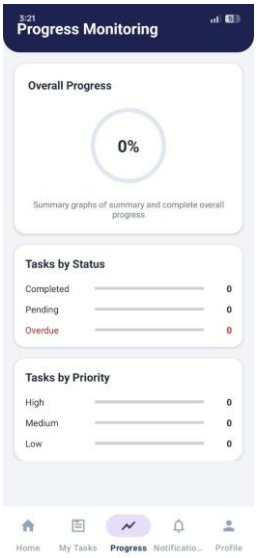
VI. Appendices

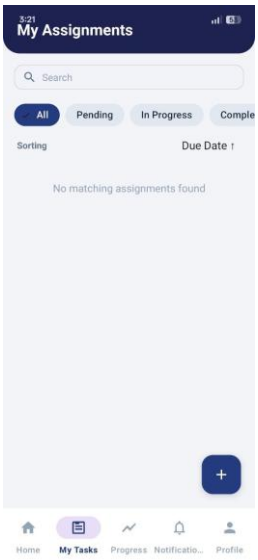
Additional screenshots

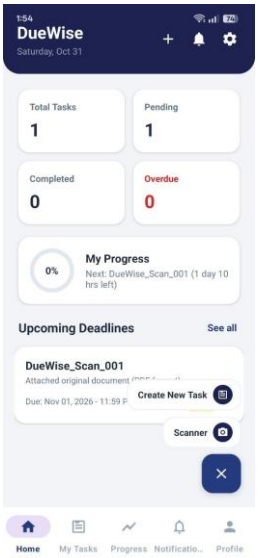
This section contains screenshots of the DueWise application to show its actual appearance and features. These screenshots serve as visual references for the different parts of the application and show how users interact with its interface. They also provide a clearer view of the application's design and how its features work.

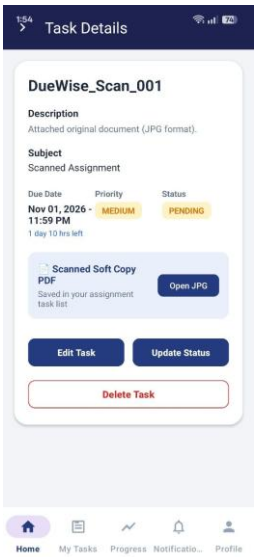












GitHub repository

https://github.com/cedricksalvador3-art/Assignment-Task-Management-App?fbclid=IwY2xjawVSaRdleHRuA2FlbQIxMQBwZG9mA3NydGMGYXBwX2lkATAAAR9KR5Ju8K5vMdVF3w2YK60zGwQZfIffWL-nTwTm-ZKkVrS6WSR9SVqq2VMgbA_aem_vx2j2WTmTXUav2v0gYTEgg

